“Data-Driven Analysis of Business Performance using Correlation and Visualization Techniques”

A PROJECT SUBMITTED TO

SYNTECXHUB Technologies

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FOR THE INTERNSHIP

IN

DATA SCIENCE

By

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Introduction

In today's data-driven business environment, analyzing relationships between key variables is essential for making informed decisions. This project focuses on exploring and understanding the relationships between important business factors such as advertising spend, sales, profit and discount.

Using Python and data analysis libraries like Pandas, Seaborn and Matplotlib, the dataset is processed and visualized to identify patterns and trends. Correlation analysis is performed to measure the strength and direction of relationships between numerical variables, while visualization techniques such as heatmaps and pairplots are used to present these relationships clearly.

The project also aims to derive meaningful insights that can support business decision making. By understanding how factors like advertising and discount impact sales and profit, organizations can optimize their strategies to improve overall performance and profitability.

Overall, This project focuses on analyzing relationships between key business variables such as sales, advertising, profit and discount using correlation analysis and data visualization techniques. The goal is to extract meaningful insights to support data-driven decision-making.

Objectives

- To load and select important numerical variables using Pandas.
- To analyze the relationship between sales and advertising spend.
- To evaluate the impact of discounts on profit.
- To compute and visualize correlations between numerical variables using a heatmap.
- To analyze pairwise relationships using pairplot.
- To identify strong positive and negative relationships among variables.

Methodology

The methodology of this project follows a structured approach to analyze the advertising dataset using the Pandas library in Python.

Data Source:

The dataset used for this project is a Sales and Marketing Dataset, obtained from Kaggle. It contains 200 rows and 5 columns with information related to advertising spend, sales, profit, discount and regional performance. The dataset consists of multiple observations with both numerical and categorical variables, allowing detailed analysis of business performance.

Variables Description:

- Advertising Spend: Amount invested in marketing and promotional activities
- Sales: Total revenue generated from sales
- Profit: Net profit earned after expenses
- Discount: Percentage of discount offered on products
- Region: Geographic region where sales are generated

Process:

First, the dataset was loaded using the Pandas library and initial checks such as viewing the first few rows, data types and missing values were performed to understand the structure of the data.

Next, important numerical variables such as advertising spend, sales, profit and discount were selected for analysis using appropriate data filtering techniques.

After that, correlation analysis was performed to identify relationships between numerical variables. The correlation matrix was visualized using a heatmap to clearly understand the strength and direction of relationships.

Further, a pairplot was created to analyze pairwise relationships and observe patterns and trends between variables.

Finally, strong positive and negative correlations were identified and meaningful insights were derived to understand how advertising and discount strategies impact sales and profit.

Data Analysis

- ❖ **To load and select important numerical variables using Pandas.**

Code:

```
import pandas as pd

# Step 1: Load dataset
df = pd.read_csv("C:\\Users\\Dell\\Downloads\\sales_dataset.csv")

# Step 2: View first few rows
print(df.head())

# Step 3: Check data types
print(df.info())

# Step 4: Select only numerical columns
numeric_df = df.select_dtypes(include=['number'])

# Step 5: Display selected numerical data
print(numeric_df.head())

# Step 6: Check columns selected
print("Numerical Columns:", numeric_df.columns)
```

Output:

Data columns (total 5 columns):

	Column	Non-Null Count	Dtype
0	Advertising_Spend	200 non-null	int64
1	Discount	200 non-null	Float64
2	Sales	200 non-null	int64
3	Profit	200 non-null	int64
4	Region	200 non-null	object

dtypes: float64(1), int64(3), object(1)

memory usage: 7.9+ KB

Selected Columns:

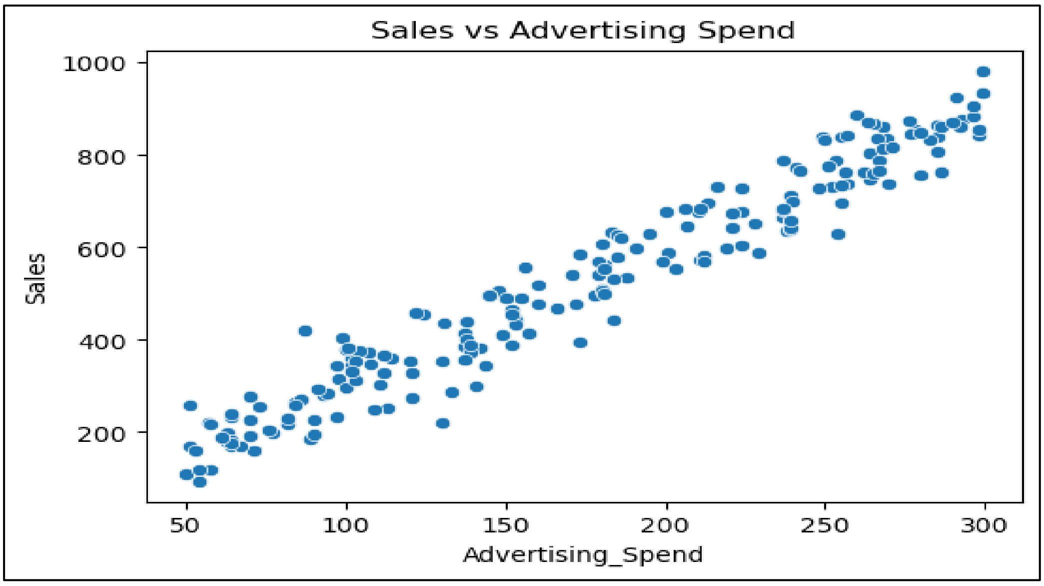
	Advertising_Spend	Discount	Sales	Profit
46	152	0.069666	390	114
55	229	0.302209	587	167
58	142	0.269921	383	86
34	64	0.101531	234	46
88	156	0.471427	557	79

- **To analyze the relationship between sales and advertising spend.**

Code:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("C:\\Users\\Dell\\Downloads\\sales_dataset.csv")

plt.figure(figsize=(6,4))
sns.scatterplot(x="Advertising_Spend", y="Sales", data=df)
plt.title("Sales vs Advertising Spend")
plt.savefig("sales_vs_advertising.png")
plt.show()

# Correlation value
print("Correlation:", df["Sales"].corr(df["Advertising_Spend"]))
```

Output:



Correlation: 0.9721

Interpretation

The scatter plot shows a strong positive relationship between advertising spend and sales. As advertising spend increases, sales also increase consistently, forming a clear upward trend.

Conclusion

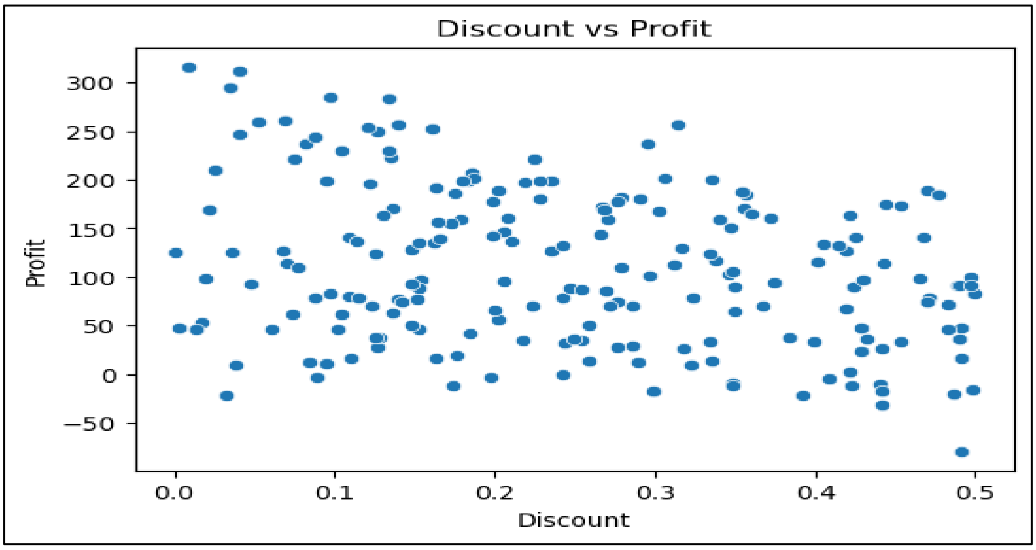
It indicate that increased marketing efforts lead to higher sales. This suggests that optimizing advertising budget can be an effective strategy to maximize sales performance.

❖ **To evaluate the impact of discounts on profit.**

Code:

```
plt.figure(figsize=(6,4))

sns.scatterplot(x="Discount", y="Profit", data=df)

plt.title("Discount vs Profit")

plt.savefig("discount_vs_profit.png")

plt.show()


# Correlation

print("Correlation:", df["Discount"].corr(df["Profit"]))
```

Output:



Correlation: -0.2985

Interpretation

The scatter plot and correlation value (-0.298) indicate a moderate negative relationship between discount and profit. As the discount increases, profit tends to decrease. This suggests that higher discounts reduce overall profitability, as lowering prices impacts the profit margin.

Conclusion

This highlights the importance of optimizing discount strategies to maintain a balance between sales growth and profitability.

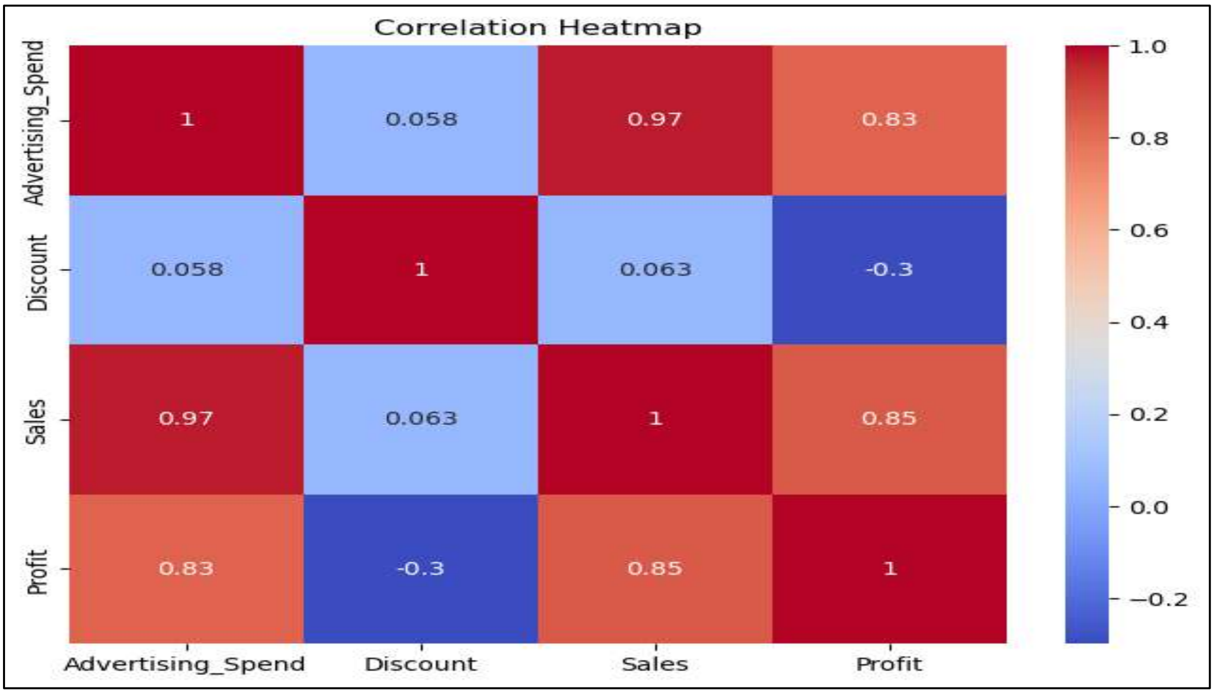
- **To compute and visualize correlations between numerical variables using a heatmap.**

Code:

```
import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt


# Load dataset

df = pd.read_csv("C:\\Users\\Dell\\Downloads\\sales_dataset.csv")


# Select only numerical columns

numeric_df = df.select_dtypes(include=['number'])
```

```
# Compute correlation
corr_matrix = numeric_df.corr()

# Plot heatmap
plt.figure(figsize=(8,6))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.savefig("heatmap.png")
plt.show()
```

Output:



Interpretation

The heatmap shows several strong relationships between variables. Advertising spend has a very strong positive correlation with sales (0.97), indicating that increased marketing significantly boosts sales. Sales and profit also show a strong positive correlation (0.85), meaning higher sales lead to higher profit.

Discount has a moderate negative correlation with profit (-0.3), suggesting that higher discounts reduce profitability.

Conclusion

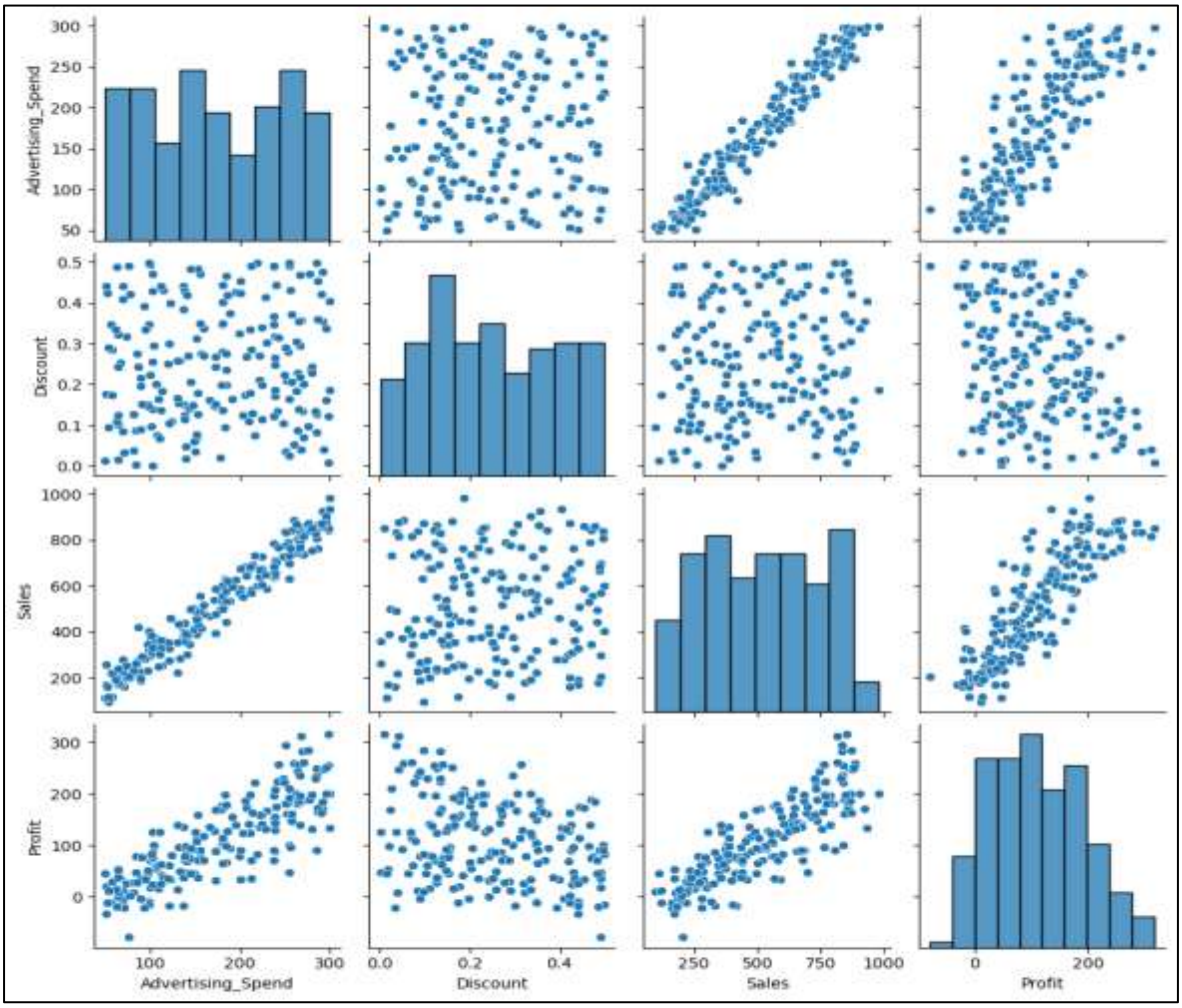
Advertising strongly drives sales and profit, while higher discounts reduce profitability. This highlights the importance of balancing marketing investment and discount strategies for optimal business performance.

- **To analyze pairwise relationships using pairplot.**

Code:

```
# Pairplot for all numerical variables
sns.pairplot(numeric_df)


plt.savefig("pairplot.png")
plt.show()
```

Output:



Interpretation

The pairplot shows clear relationships between key variables. There is a strong positive relationship between advertising spend and sales, as well as between sales and profit, indicating that higher marketing efforts lead to increased revenue and profitability.

On the other hand, discount does not show a strong relationship with sales, but it exhibits a slight negative trend with profit, suggesting that higher discounts may reduce profit margins.

Conclusion

The analysis indicates that advertising is a major driver of sales growth, which in turn significantly impacts profit. The strong linear patterns between advertising–sales and sales–profit confirm a consistent upward trend.

In contrast, discount appears to have limited influence on increasing sales, while it negatively affects profit. This suggests that excessive discounting may not be an effective long-term strategy.

Therefore, businesses should focus on increasing advertising efficiency to boost sales, while carefully managing discount strategies to avoid loss in profitability.

- **To identify strong positive and negative relationships among variables.**

Code:

```
# Get correlation values
corr_matrix = numeric_df.corr()

# Print strong positive and negative correlations
print("Strong Positive Correlations (> 0.5):")
print(corr_matrix[corr_matrix > 0.5])

print("\nStrong Negative Correlations (< -0.5):")
print(corr_matrix[corr_matrix < -0.5])
```

Output:

Strong Positive Correlations:

Advertising_Spend	Sales	0.972154
Profit	Sales	0.849211
Advertising_Spend	Profit	0.828591

Strong Negative Correlations:

No strong negative correlations were observed in the dataset.

Interpretation

The analysis of correlation values shows that there are several strong positive relationships among the variables. Advertising spend and sales exhibit a very strong positive correlation (0.97), indicating that increased advertising significantly drives sales.

Sales and profit have a strong positive correlation (0.85), suggesting that higher sales directly contribute to increased profitability. Advertising spend and profit also show a strong positive relationship (0.83).

Conclusion

These results indicate that advertising plays a crucial role in business growth, as it directly influences sales, which in turn impacts profit. The strong positive relationships highlight a clear dependency between marketing efforts and overall performance.

Although no strong negative correlations were found, a moderate negative relationship between discount and profit exists, suggesting that higher discounts may reduce profit margins.

Therefore, businesses should focus on increasing advertising efficiency while carefully managing discount strategies to maintain profitability.

Overall Conclusion

This project successfully analyzed the relationships between key business variables such as advertising spend, sales, profit and discount using data visualization and correlation techniques.

The analysis revealed that advertising spend has a strong positive impact on sales, indicating that increased marketing efforts significantly contribute to higher revenue. Furthermore, sales and profit were also found to be strongly positively correlated, suggesting that an increase in sales directly leads to improved profitability.

On the other hand, discount showed a moderate negative relationship with profit, highlighting that excessive discounting can reduce profit margins. No strong negative correlations were observed among the variables, indicating that most factors positively contribute to business performance.

Overall, the findings emphasize the importance of data-driven decision-making, where businesses can optimize advertising strategies to boost sales while carefully managing discount policies to maintain profitability.